

MOHNISH RAMACHANDRAN

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OBJECTIVE

AI/ML Engineer specializing in Computer Vision, LLMs, and Generative AI, with hands-on experience developing, evaluating, and deploying machine learning models. Adept at transforming complex datasets into actionable insights and building AI systems that deliver measurable business and product impact.

EDUCATION

Master of Computer Science, North Carolina State University Aug 2025 - Present
Coursework: *Foundations of Data Science, Automated Learning & Data Analysis, Artificial Intelligence*
CGPA: 3.45/4.00

Bachelor of Technology, SRM Institute of Science and Technology, Chennai, India 2021 - 2025
Degree: B.Tech in Artificial Intelligence CGPA: 9.14/10.00
Coursework: *Neural Networks & Machine Learning, Image & Video Processing, Reinforcement Learning*

SKILLS

Programming	Python, SQL, C++
Data Science	Data Cleaning, EDA, Feature Engineering, Statistical Analysis, Hypothesis Testing, Model Evaluation
Machine Learning	Supervised & Unsupervised Learning, Regression, Classification, Clustering, Ensemble Methods
Deep Learning	CNNs, Vision Transformers, Model Benchmarking
Frameworks	Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow
Visualization	Looker Studio, Data Dashboards
MLOps & Dev	Git, Docker, ONNX

EXPERIENCE

Research Fellowship Feb 2024 - June 2024
HyperVerge Inc. India

- Project: *Qualitative and Quantitative Analysis of Deepfakes*
- Conducted systematic qualitative and quantitative evaluation of deepfake generation and detection systems using GANs and autoencoders, with a focus on model robustness and generalization.
- Implemented and benchmarked state-of-the-art detection models, analyzing failure cases and performance trade-offs across datasets.

Data Analyst Intern Jun 2023 - July 2023
Olimind Inc. San Francisco, CA

- Analyzed large-scale user interaction data to extract behavioral patterns and actionable insights for product decision-making.
- Built interactive dashboards in Looker Studio to support real-time analytics, improving mentor visibility, and reducing manual reporting effort.

PROJECTS

Crime Analysis of Raleigh, NC Performed an in-depth data analysis of publicly available government crime data for Raleigh, NC, examining geographic, temporal, district-level, and crime-type patterns to extract actionable insights.

Cancer Detection Designed an ensemble framework for breast cancer classification by integrating CNNs (VGG-16, InceptionResNetV2, ResNet-50, Xception) with a fine-tuned Hugging Face ViT.

Arrow Object Detection Developed an arrow detection module for rover navigation using SSD-MobileNet and YOLOv5; exported trained models to ONNX and managed the complete pipeline from data collection or rocessing to deployment.

AI Travel Agent Built an autonomous LLM-powered agent using RAG and tool-calling to plan personalized trips, dynamically query external APIs, reason over retrieved data, and adapt responses based on user constraints.

Deep Research Agent Designed an Agentic AI system capable of multi-step reasoning, self-critique, and iterative refinement of outputs by orchestrating LLMs with web search tool

Music Recommendation Using Sentiment Analysis Built a sentiment-aware music recommendation system by applying CNN-based text classification on user input.

CERTIFICATIONS

Deep Learning Specialization (5 courses)	DeepLearning.AI
Machine Learning Foundations	AWS Academy
Cloud Foundations	AWS Academy
Python101 for Data Science	CognitiveClass.ai

LEADERSHIP AND ACTIVITIES

- Led a research team during of 6, at an industry fellowship program by HyperVerge, coordinating experimentation, analysis, and communication with senior researchers.
- Mentor during my second year at RUDRA-Mars Rover Team (College team) for the incoming developer team.
- International Rover Challenge (IRC-24): Achieved 2nd Runners-up
- International Rover Drone Challenge (IRDC-24): Achieved 2nd Runners-up